

A Survey on IoT-Based Underground Cable Fault Detection Systems (2021–2025)

Abstract

Underground power cables are increasingly used in modern power distribution systems due to their improved safety, reliability, and reduced environmental exposure. However, detecting and locating faults in underground cables is a complex task because the cables are buried beneath the ground and are not visually accessible. Traditional fault detection techniques are often time-consuming, costly, and inefficient. With the advancement of Internet of Things (IoT) technology, real-time monitoring and intelligent fault detection systems have emerged as effective solutions. This survey paper reviews and analyzes five research papers published between 2021 and 2025 that focus on IoT-based underground cable fault detection systems. The survey compares methodologies, sensor technologies, communication platforms, datasets, challenges, and future directions, providing a comprehensive understanding of the current research landscape.

1. Background

Underground cables play a vital role in modern power transmission and distribution networks, especially in urban areas where overhead lines are impractical. Compared to overhead cables, underground cables offer advantages such as improved safety, reduced visual pollution, and protection from environmental hazards. Despite these benefits, underground cable systems face significant challenges when faults occur.

Faults in underground cables may arise due to aging, insulation failure, moisture ingress, construction activities, rodent damage, or mechanical stress. Locating these faults is difficult because the cables are hidden beneath the surface. Conventional fault detection techniques such as sectionalizing, thumping, and Murray loop methods require extensive manual effort and excavation, leading to increased downtime and repair costs. These limitations have motivated researchers to explore advanced fault detection techniques using IoT technology.

2. Purpose of the Survey

The primary purpose of this survey is to:

- Review recent research (2021–2025) on IoT-based underground cable fault detection systems
- Analyze different methodologies and architectures proposed in the literature
- Compare sensors, IoT platforms, and fault localization techniques
- Identify challenges and research gaps
- Highlight future research directions in this domain

This survey aims to serve as a reference for students, researchers, and engineers working on smart power distribution systems.

3. Overview of IoT-Based Cable Fault Detection

IoT-based underground cable fault detection systems typically consist of:

- **Sensors** (voltage, current, temperature, impedance)
- **Microcontrollers** (Arduino, PIC, NodeMCU)
- **Communication modules** (Wi-Fi, GSM, GPS)
- **Cloud platforms** (ThingSpeak, IoT dashboards)
- **Display/Alert systems** (LCD, SMS, email alerts)

These components work together to continuously monitor cable parameters, detect abnormal conditions, and notify maintenance personnel in real time.

4. Review of Selected Research Papers

4.1 Research Paper (2021)

The 2021 study proposes an IoT-based underground fault detector using voltage divider and resistance measurement principles. The system estimates fault distance by analyzing voltage variations across resistors placed along the cable. A microcontroller processes the data and displays fault location on an LCD while transmitting information via IoT. This approach is cost-effective and simple but mainly focuses on short-circuit faults and lacks advanced analytics.

4.2 Research Paper (2022)

The 2022 research introduces a multi-sensor IoT-based monitoring system that measures current and temperature parameters. The collected data is transmitted to a cloud platform for real-time monitoring. This system improves fault detection accuracy and supports preventive maintenance. However, the system heavily depends on internet connectivity and does not address cybersecurity concerns.

4.3 Research Paper (2023)

The 2023 study emphasizes wireless sensor networks combined with IoT for faster fault detection. The system detects abnormalities in electrical parameters and sends alerts to maintenance teams. This approach reduces fault response time and improves reliability. However, scalability and deployment complexity remain major challenges.

4.4 Research Paper (2024)

The 2024 paper presents an IoT-based underground cable fault detection system using Arduino, GSM, GPS, and the ThingSpeak cloud platform. The system accurately identifies fault location and sends real-time alerts to repair personnel. While the solution is effective and practical, it relies on simulated cable setups rather than real underground installations.

4.5 Research Paper (2025)

The 2025 research introduces a smart underground fault monitoring system integrating IoT with predictive analytics and MATLAB simulations. The system enables proactive maintenance and enhances grid resilience. It supports real-time visualization and future AI integration. However, implementation complexity and cost may limit widespread adoption.

5. Dataset Understanding

The datasets used in the reviewed research papers are primarily sensor-generated datasets. These datasets include:

- Voltage levels
- Current flow
- Temperature variations
- Resistance and impedance values
- Fault distance and location
- Time-stamped real-time readings

The data is collected either through real-time IoT sensors or generated using simulation tools such as MATLAB. Cloud platforms store and visualize this data, enabling fault analysis, system evaluation, and decision-making. These datasets are essential for detecting faults, estimating fault locations, and improving system reliability.

6. Comparative Analysis

Year	Methodology	Sensors Used	IoT Platform	Strengths	Limitations
2021	Voltage divider method	Voltage	IoT cloud	Simple, low cost	Limited fault types
2022	Multi-sensor monitoring	Current, Temp	Cloud IoT	Improved accuracy	Internet dependency
2023	Wireless sensor network	Electrical sensors	IoT gateway	Fast alerts	Scalability issues
2024	Arduino + GPS + GSM	Voltage, Current	ThingSpeak	Exact fault location	Simulated setup
2025	IoT + predictive analytics	Multi-parameter	Advanced cloud	Proactive monitoring	High complexity

7. Challenges

Despite advancements, IoT-based underground cable fault detection systems face several challenges:

- High initial installation cost
- Dependence on continuous internet connectivity
- Limited real-world deployment
- Cybersecurity and data privacy risks
- Lack of standardized datasets

8. Research Gaps

The survey identifies the following research gaps:

- Limited use of machine learning for predictive fault detection
- Insufficient real-time datasets from actual underground cable installations
- Lack of standard benchmarking techniques
- Minimal focus on cybersecurity in IoT-based systems

9. Correlation and Inferences

The reviewed studies show a strong correlation between real-time sensor data and fault detection accuracy. Multi-sensor systems provide better fault diagnosis compared to single-parameter systems. Cloud-based monitoring significantly reduces fault response time. It can be inferred that integrating IoT with advanced analytics improves system reliability and maintenance efficiency.

10. Future Directions

Future research should focus on:

- Integrating AI and machine learning for predictive maintenance
- Developing standardized and open datasets
- Enhancing cybersecurity frameworks
- Implementing self-healing and autonomous fault recovery systems
- Deploying large-scale real-world IoT-based fault detection systems

11. Conclusion

This survey reviewed five research papers published between 2021 and 2025 on IoT-based underground cable fault detection systems. The analysis demonstrates that IoT technology significantly enhances fault detection accuracy, reduces downtime, and enables real-time monitoring. Although challenges remain, IoT-based solutions.

References

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